

Cross-Section Comparison: RS67 vs RS57-70

Analysis Report

April 28, 2026

Live Protons on Target (POT) Summary

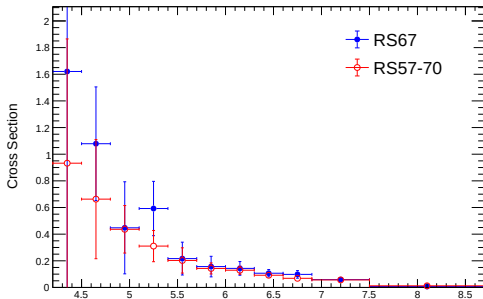
Roadset	LH2 POT	Flask POT	LD2 POT	Total (H2+D2+F)
57	3.79×10^{16}	4.23×10^{15}	1.90×10^{16}	6.11×10^{16}
59	8.44×10^{15}	9.18×10^{14}	3.84×10^{15}	1.32×10^{16}
62	5.51×10^{16}	1.13×10^{16}	2.55×10^{16}	9.19×10^{16}
67	1.57×10^{17}	3.58×10^{16}	7.51×10^{16}	2.68×10^{17}
70	1.76×10^{16}	3.71×10^{15}	8.54×10^{15}	2.99×10^{16}
57–70 Combined	2.76×10^{17}	5.59×10^{16}	1.32×10^{17}	4.64×10^{17}

- POT values extracted from analysis header file.
- RS57–70 used as the primary integrated dataset for comparison.

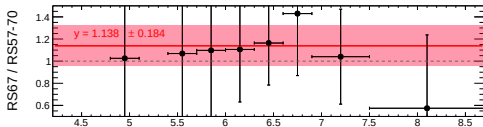
Comparison for $0.0 \leq x_F < 0.05$

Target: LH2

LH2 Cross Section, $0.0 \leq x_F < 0.05$

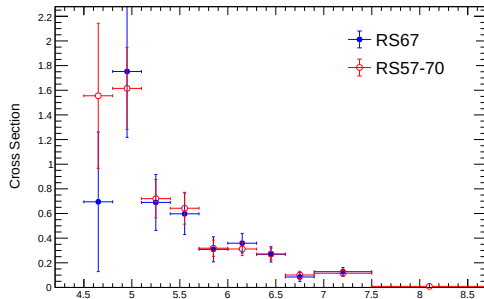


Invariant Mass (GeV)

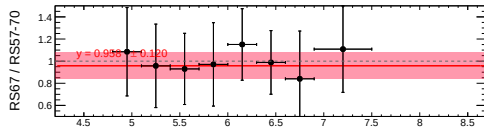


Target: LD2

LD2 Cross Section, $0.0 \leq x_F < 0.05$



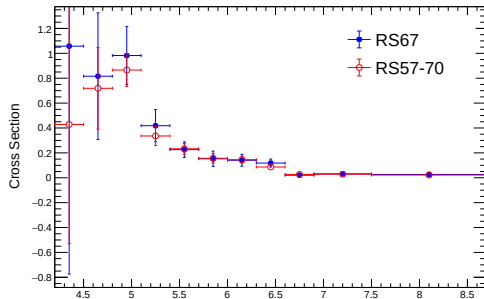
Invariant Mass (GeV)



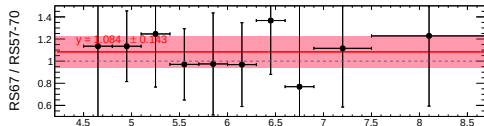
Comparison for $0.05 \leq x_F < 0.1$

Target: LH2

LH2 Cross Section, $0.05 \leq x_F < 0.1$

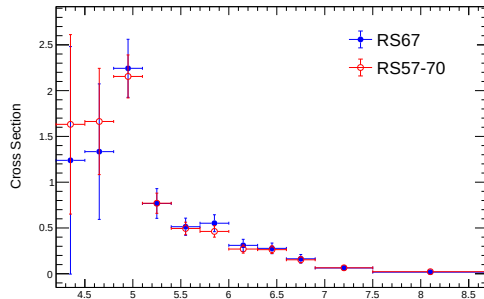


Invariant Mass (GeV)

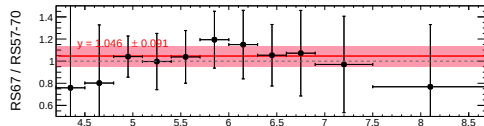


Target: LD2

LD2 Cross Section, $0.05 \leq x_F < 0.1$



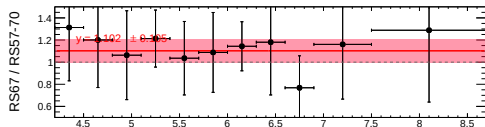
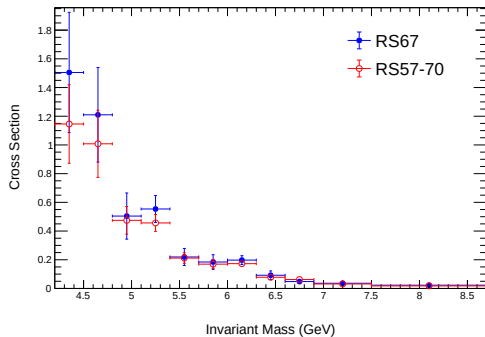
Invariant Mass (GeV)



Comparison for $0.1 \leq x_F < 0.15$

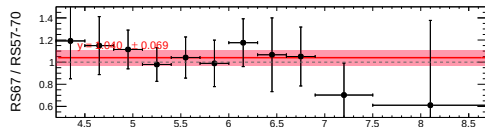
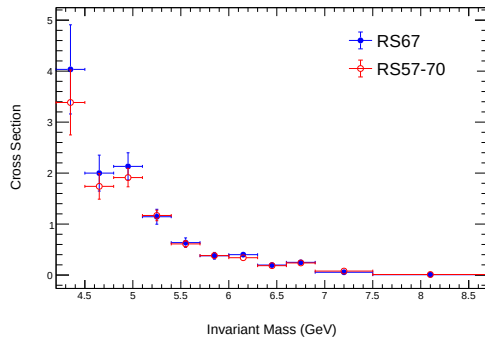
Target: LH2

LH2 Cross Section, $0.1 \leq x_F < 0.15$



Target: LD2

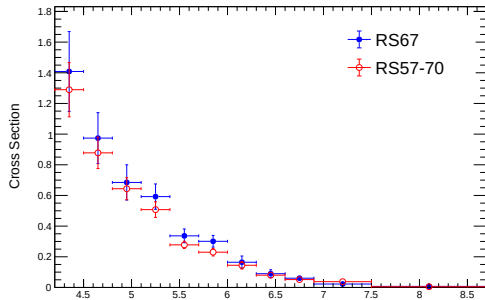
LD2 Cross Section, $0.1 \leq x_F < 0.15$



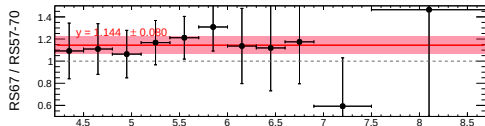
Comparison for $0.15 \leq x_F < 0.2$

Target: LH2

LH2 Cross Section, $0.15 \leq x_F < 0.2$

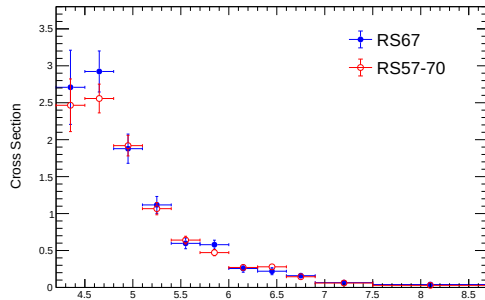


Invariant Mass (GeV)

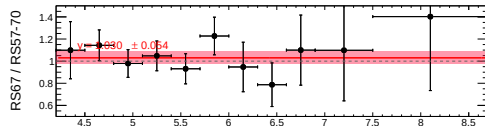


Target: LD2

LD2 Cross Section, $0.15 \leq x_F < 0.2$



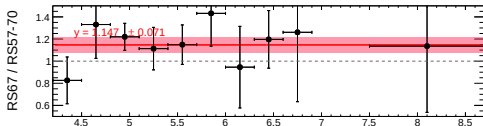
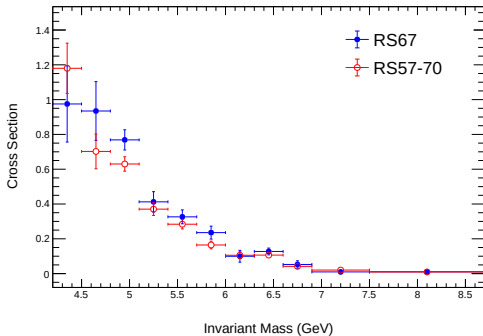
Invariant Mass (GeV)



Comparison for $0.2 \leq x_F < 0.25$

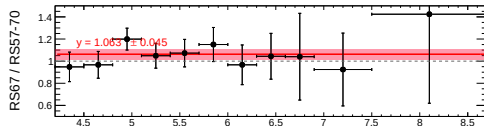
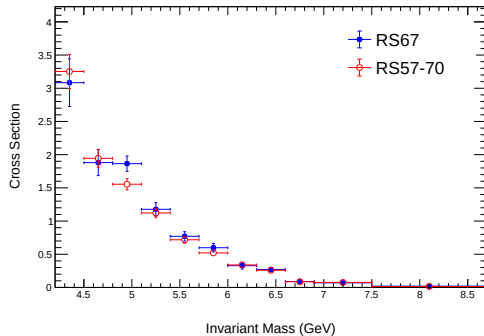
Target: LH2

LH2 Cross Section, $0.2 \leq x_F < 0.25$



Target: LD2

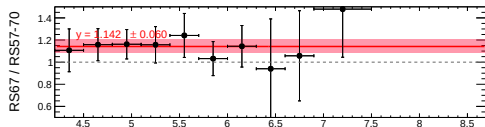
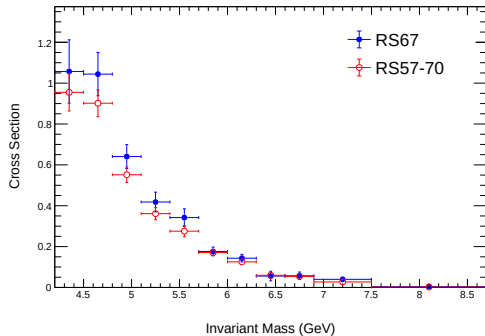
LD2 Cross Section, $0.2 \leq x_F < 0.25$



Comparison for $0.25 \leq x_F < 0.3$

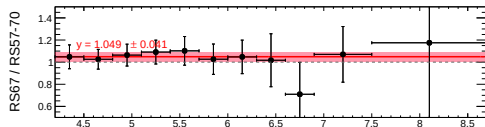
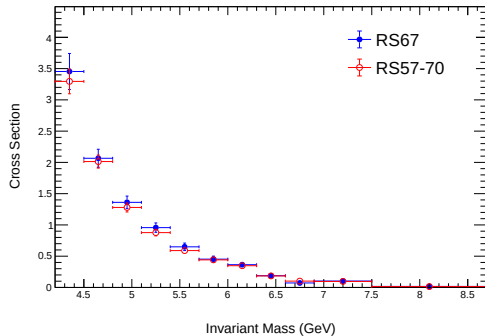
Target: LH2

LH2 Cross Section, $0.25 \leq x_F < 0.3$



Target: LD2

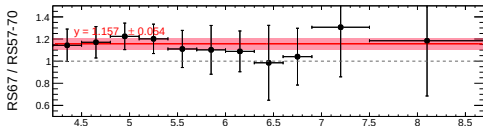
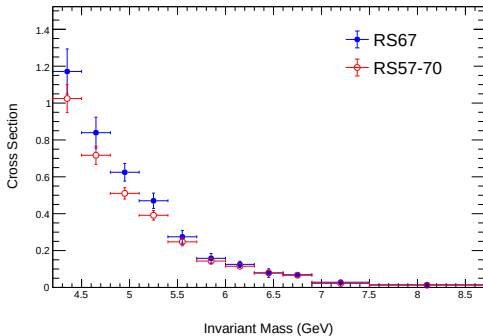
LD2 Cross Section, $0.25 \leq x_F < 0.3$



Comparison for $0.3 \leq x_F < 0.35$

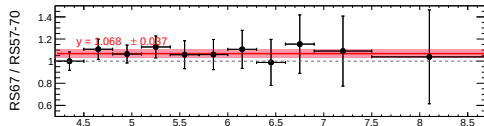
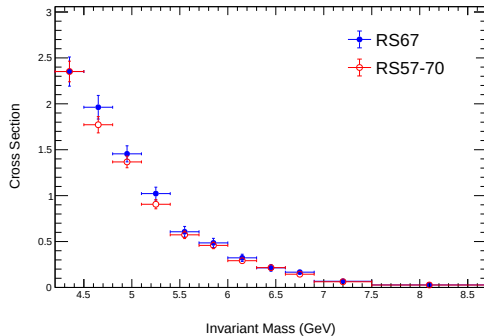
Target: LH2

LH2 Cross Section, $0.3 \leq x_F < 0.35$



Target: LD2

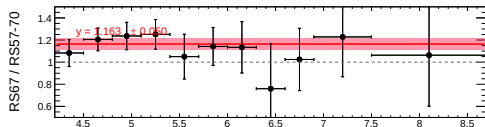
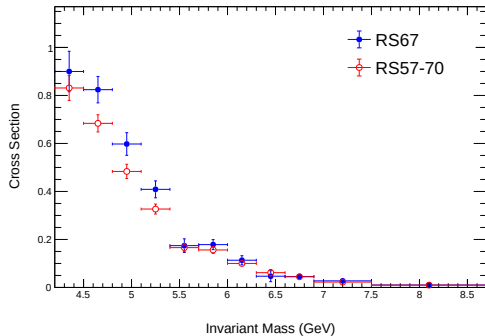
LD2 Cross Section, $0.3 \leq x_F < 0.35$



Comparison for $0.35 \leq x_F < 0.4$

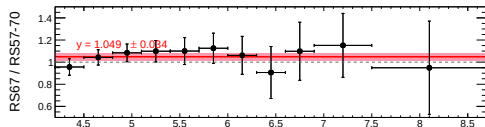
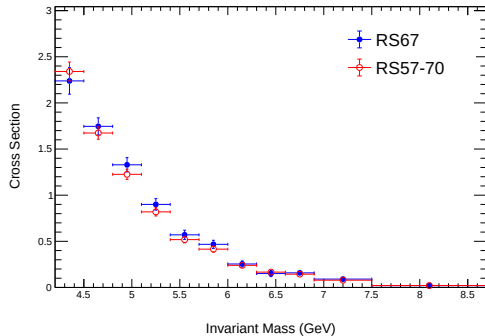
Target: LH2

LH2 Cross Section, $0.35 \leq x_F < 0.4$



Target: LD2

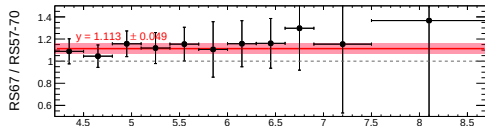
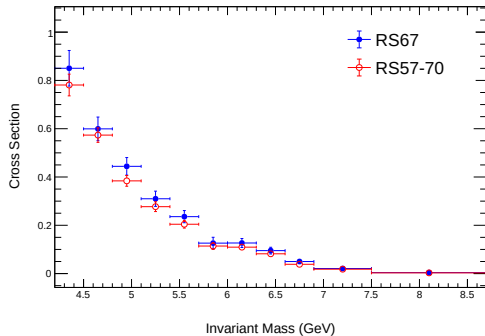
LD2 Cross Section, $0.35 \leq x_F < 0.4$



Comparison for $0.4 \leq x_F < 0.45$

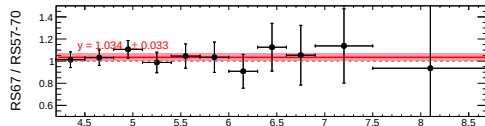
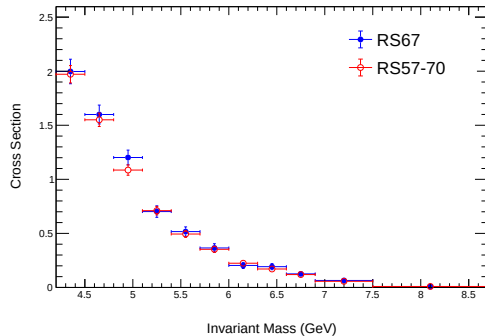
Target: LH2

LH2 Cross Section, $0.4 \leq x_F < 0.45$



Target: LD2

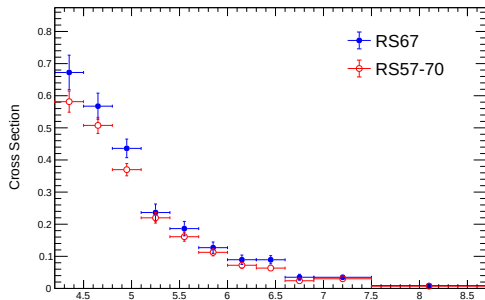
LD2 Cross Section, $0.4 \leq x_F < 0.45$



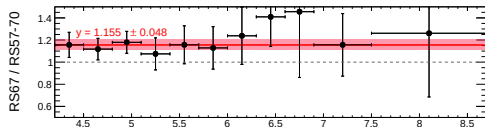
Comparison for $0.45 \leq x_F < 0.5$

Target: LH2

LH2 Cross Section, $0.45 \leq x_F < 0.5$

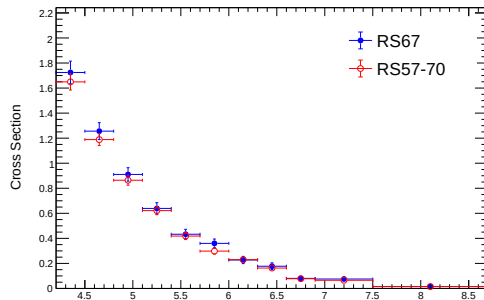


Invariant Mass (GeV)

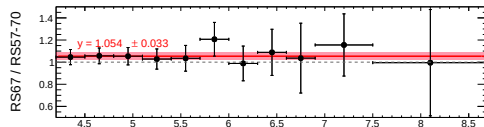


Target: LD2

LD2 Cross Section, $0.45 \leq x_F < 0.5$



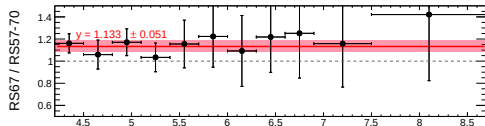
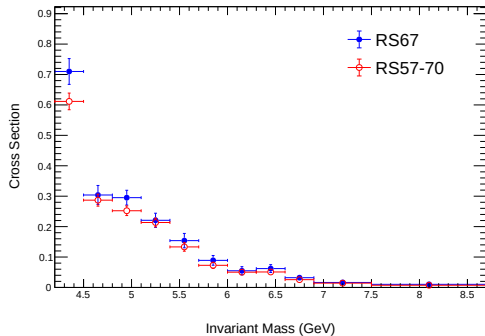
Invariant Mass (GeV)



Comparison for $0.5 \leq x_F < 0.55$

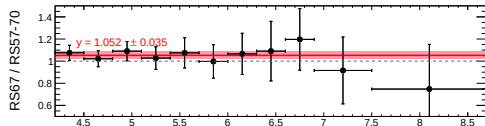
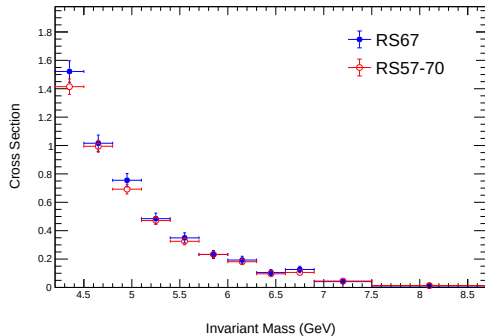
Target: LH2

LH2 Cross Section, $0.5 \leq x_F < 0.55$



Target: LD2

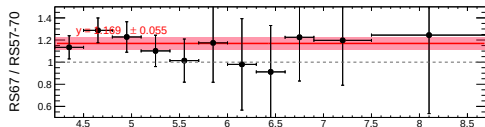
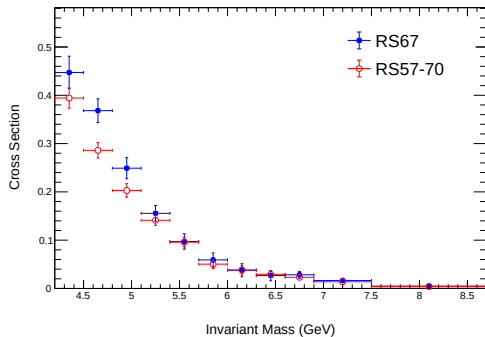
LD2 Cross Section, $0.5 \leq x_F < 0.55$



Comparison for $0.55 \leq x_F < 0.6$

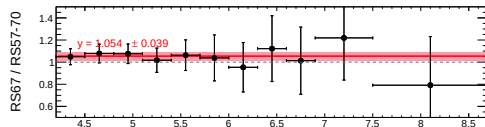
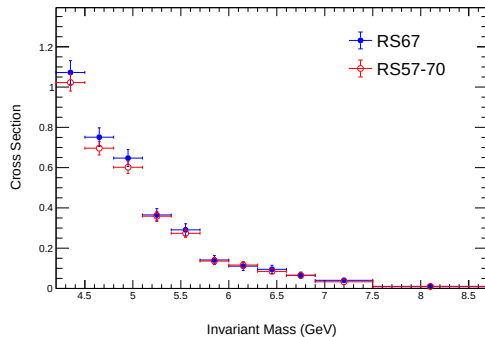
Target: LH2

LH2 Cross Section, $0.55 \leq x_F < 0.6$



Target: LD2

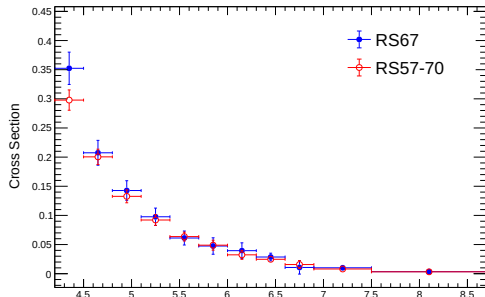
LD2 Cross Section, $0.55 \leq x_F < 0.6$



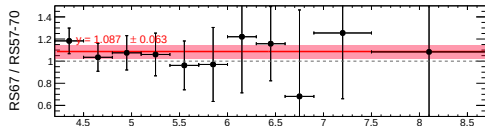
Comparison for $0.6 \leq x_F < 0.65$

Target: LH2

LH2 Cross Section, $0.6 \leq x_F < 0.65$

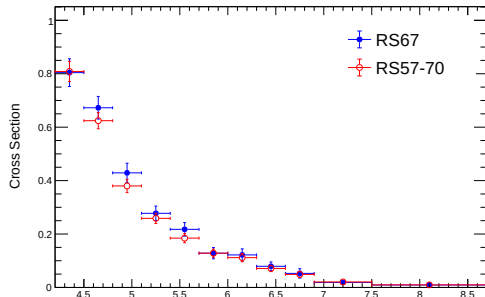


Invariant Mass (GeV)

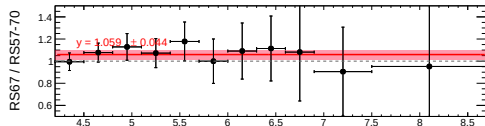


Target: LD2

LD2 Cross Section, $0.6 \leq x_F < 0.65$



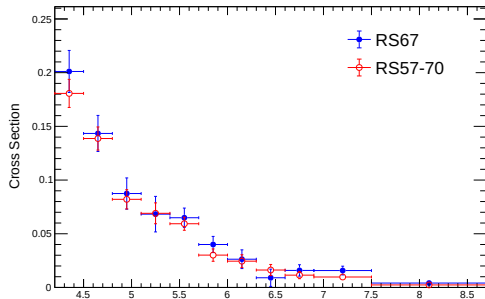
Invariant Mass (GeV)



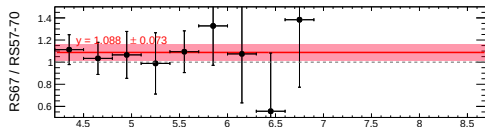
Comparison for $0.65 \leq x_F < 0.7$

Target: LH2

LH2 Cross Section, $0.65 \leq x_F < 0.7$

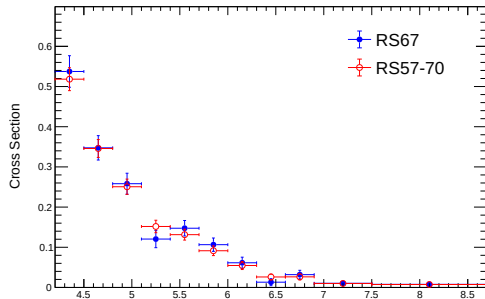


Invariant Mass (GeV)

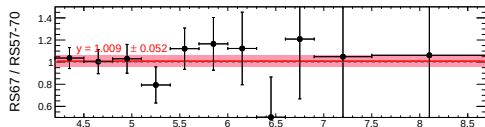


Target: LD2

LD2 Cross Section, $0.65 \leq x_F < 0.7$



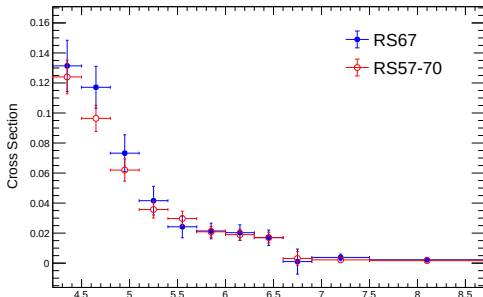
Invariant Mass (GeV)



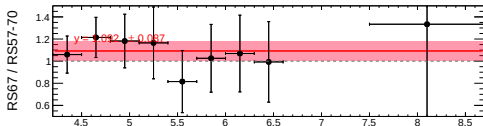
Comparison for $0.7 \leq x_F < 0.75$

Target: LH2

LH2 Cross Section, $0.7 \leq x_F < 0.75$

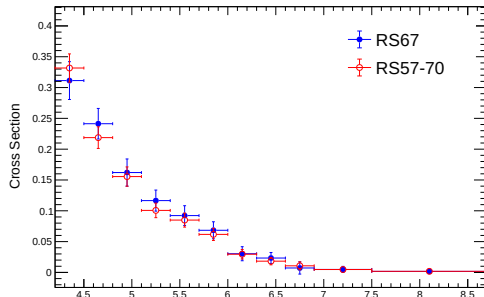


Invariant Mass (GeV)

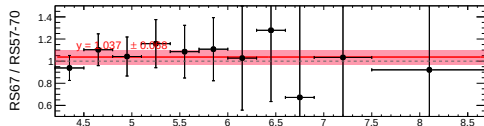


Target: LD2

LD2 Cross Section, $0.7 \leq x_F < 0.75$



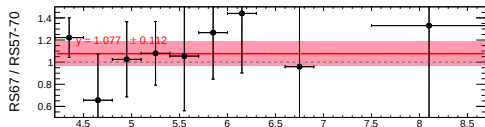
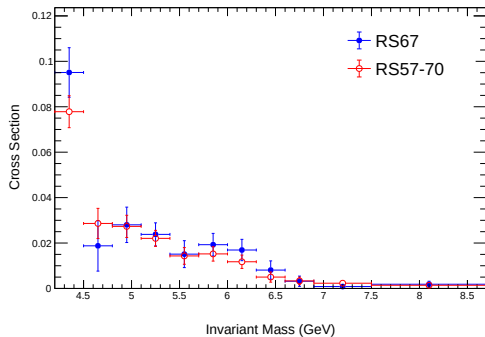
Invariant Mass (GeV)



Comparison for $0.75 \leq x_F < 0.8$

Target: LH2

LH2 Cross Section, $0.75 \leq x_F < 0.8$



Target: LD2

LD2 Cross Section, $0.75 \leq x_F < 0.8$

